ETH zürich

Beyond Model Democracy in CMIP 6

Performance- and Interdependence-Based Weighting of Future Warming Projections

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1. Motivation

Why use multi-model ensembles average?

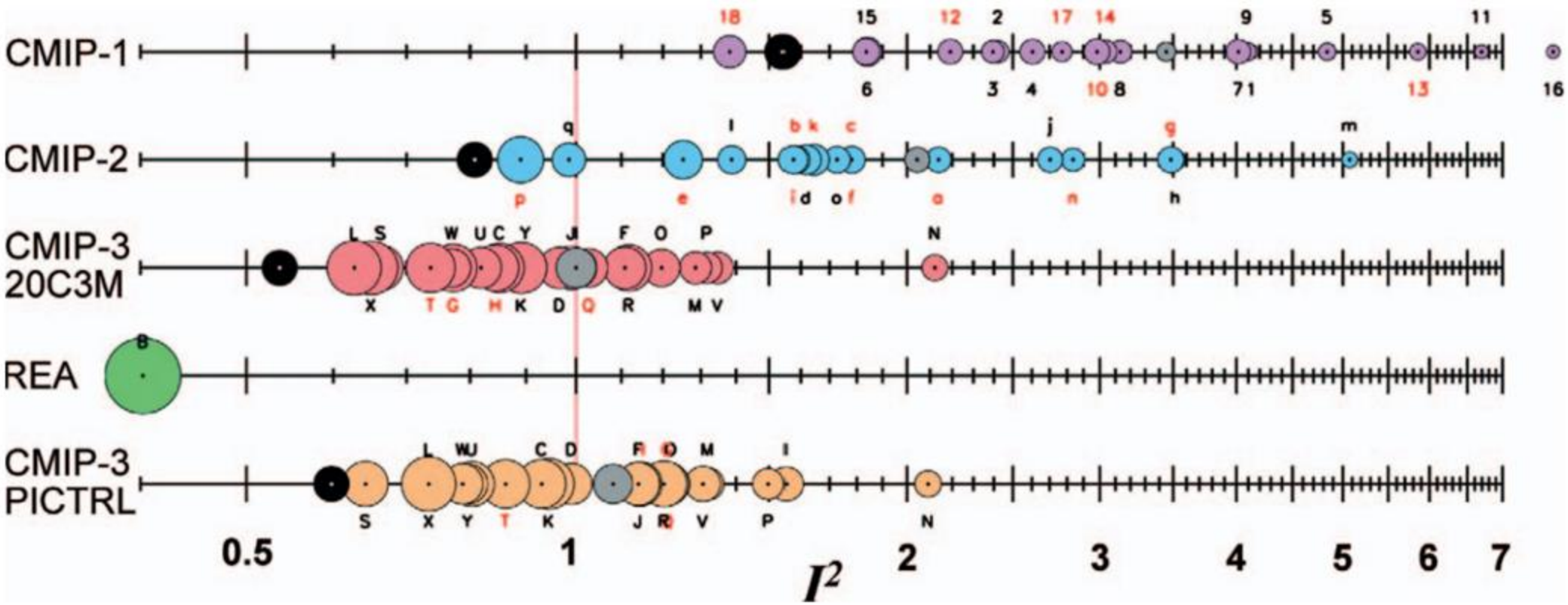


Figure 1: Performance index I² by model and generation.

Lower I² means **better agreement** with observations

2. Problem

Limitations of Model democracy

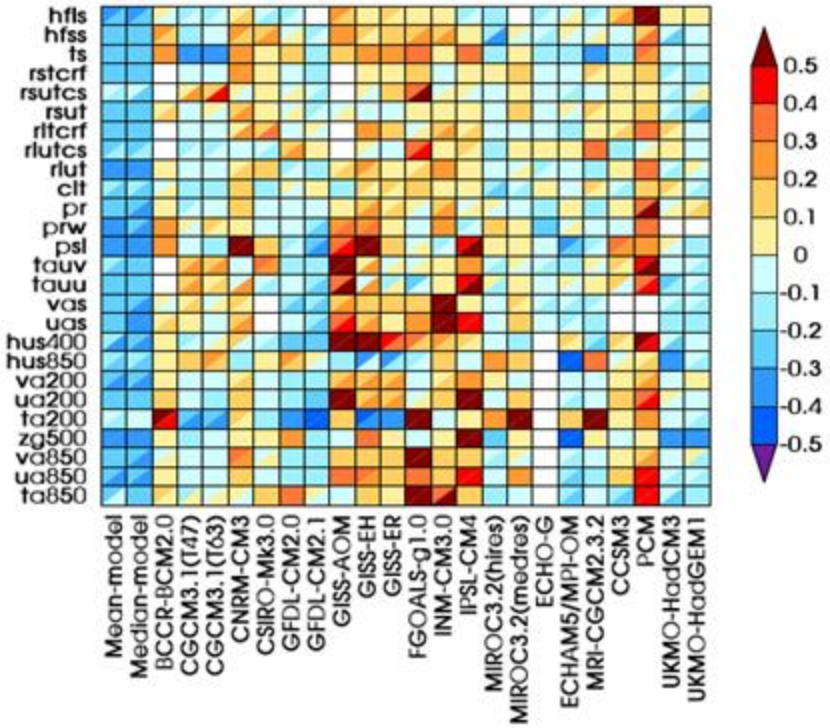
based on the assumptions:

1) reasonably independent
2) equally plausible
3) distributed around reality
4) range of projections is representative of what we believe is uncertainty in the projected quantity

! none of our four assumptions are fulfilled by model democracy !

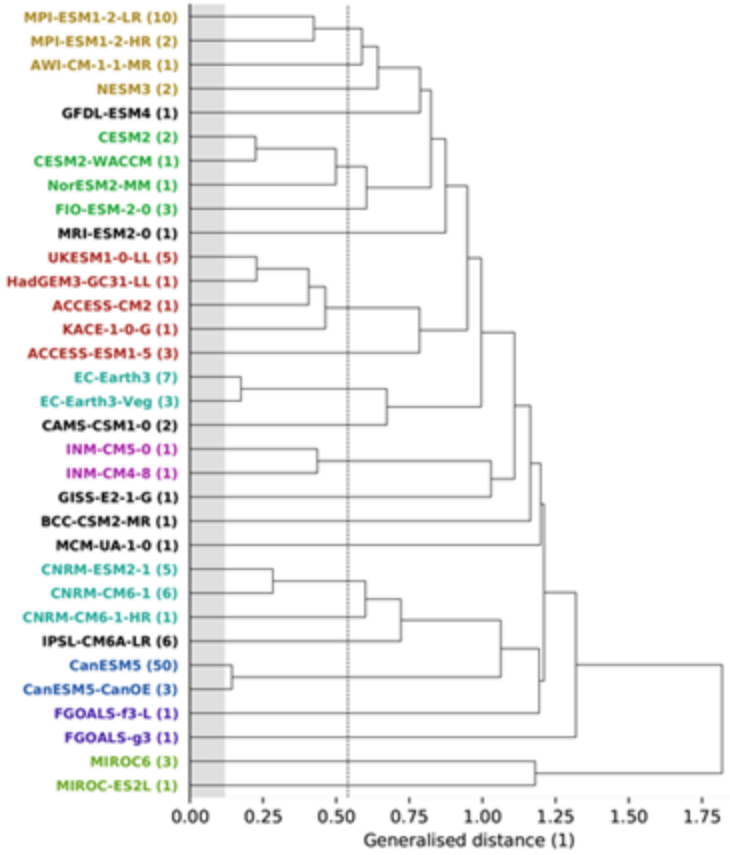
3. Evidence

Model performance and interdependence



Performance

Figure 2: Model performance and quality



Interdependence

Figure 3: Climate model genealogy: Similarity based on model biases

3. Evidence

Model performance and interdependence

5 reasons to consider model **performance** and **interdependence**:

1) massiv model spread for some region and variables
2) For base state dependant processes, projected anomalies relative to today is problematic + scaling methods & bias correction may be unfeasible
3) some cases emergent constraints can improve projections
4) large initial condition ensembles hard to combine with single runs from other models
5) model dependence increasingly relevant with replication of code

3. Evidence

Model performance and interdependence

need to consider model **performance** and **interdependence** for at least 5 reasons:

1) some region and variables the model spread is massive
2) where processes are sensitive to base state working with projected anomalies relative to today is problematic and scaling methods and bias correction may be unfeasible
3) some cases emergent constraints clearly relevant for model evaluation that can improve projections (need to understand further?)
4) large initial condition ensembles are hard to combine with single runs from other models
5) model dependence gets increasingly relevant with increasing replication of code

4. Method

Performance–interdependence weighting

Simplified:
- Models closer to observations → higher weight
- Models similar to others → lower weight

core ideas

1) models that agree poorly with observations for a set of diagnostics get less weight
2) models that largely duplicate existing models also get less weight

distance metric of model i to observation

$$w_i = e^{-\frac{D_i^2}{\sigma_D^2}} \Big/ \left(1 + \sum_{j \neq i}^{M} e^{-\frac{S_{ij}^2}{\sigma_S^2}}\right)$$

distance metric between model i and j

single model weight

determine how strongly model performance and similarity are measured

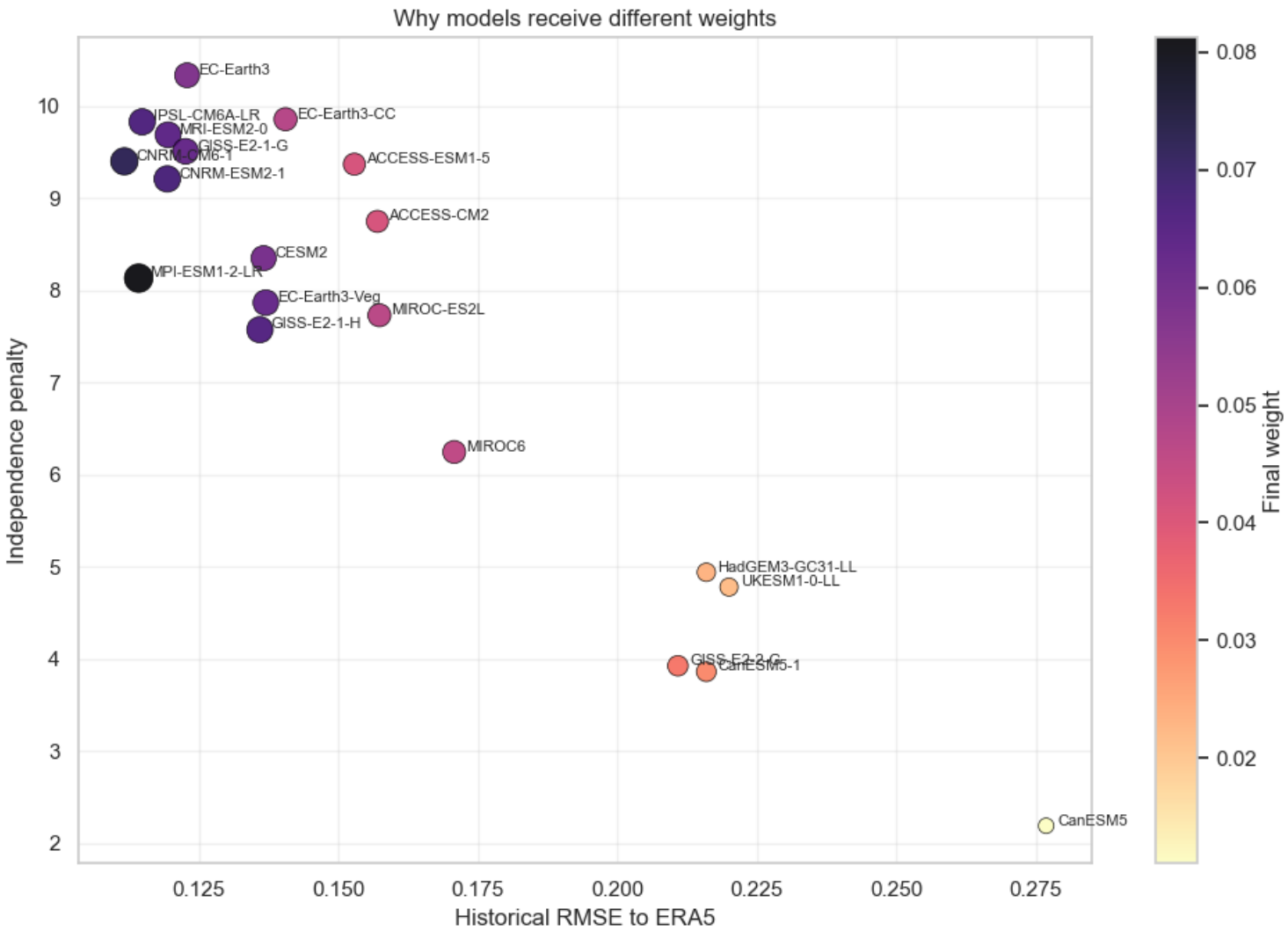
D_i and S_{ij} are chosen to be the root-mean-square difference

4. Method

Performance–interdependence weighting

4. Method Interdependence

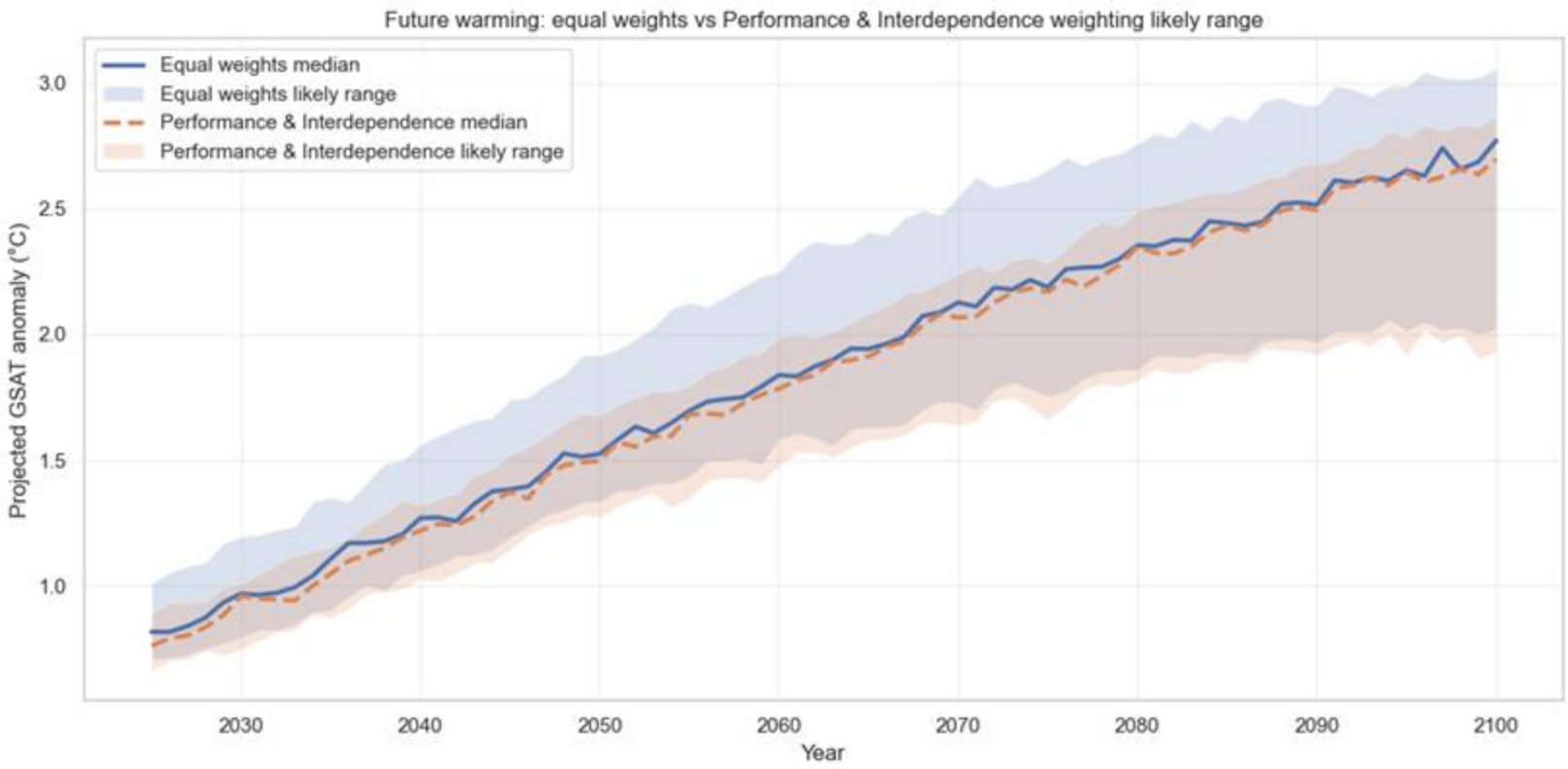
5. Results


Why models receive different weights
EC-Earth3
IPSL-CM6A-LR
EC-Earth3-CC
MRI-ESM2-0
GISS-E2-1-G
CNRM-CM6-1
ACCESS-ESM1-5
CNRM-ESM2-1
ACCESS-CM2
CESM2
MPI-ESM1-2-LR
EC-Earth3-Veg
MIROC-ES2L
GISS-E2-1-H
MIROC6
HadGEM3-GC31-LL
UKESM1-0-LL
GISS-E2-2-G
CanESM5-1
CanESM5
Independence penalty
Historical RMSE to ERA5
Final weight
10
9
8
7
6
5
4
3
2
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0.08
0.07
0.06
0.05
0.04
0.03
0.02

Hindcasting/Proof for Performance?

5. Results

Impact on Projections



1. Reduced Uncertainty Range

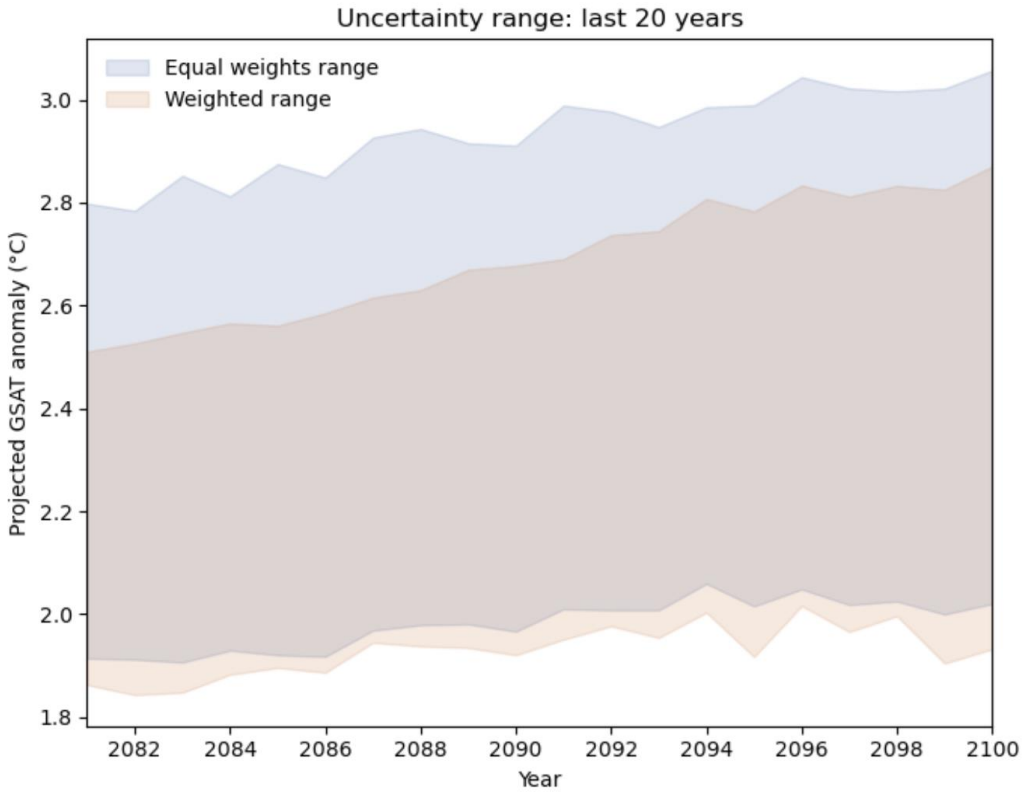
- narrower spread
- fewer outliers
- filtered model space

2. Decrease in Projected Warming

- slight downward shift
- high-RMSE models receive lower weight
- more moderate projections

5. Results

1. Reduced Uncertainty Range

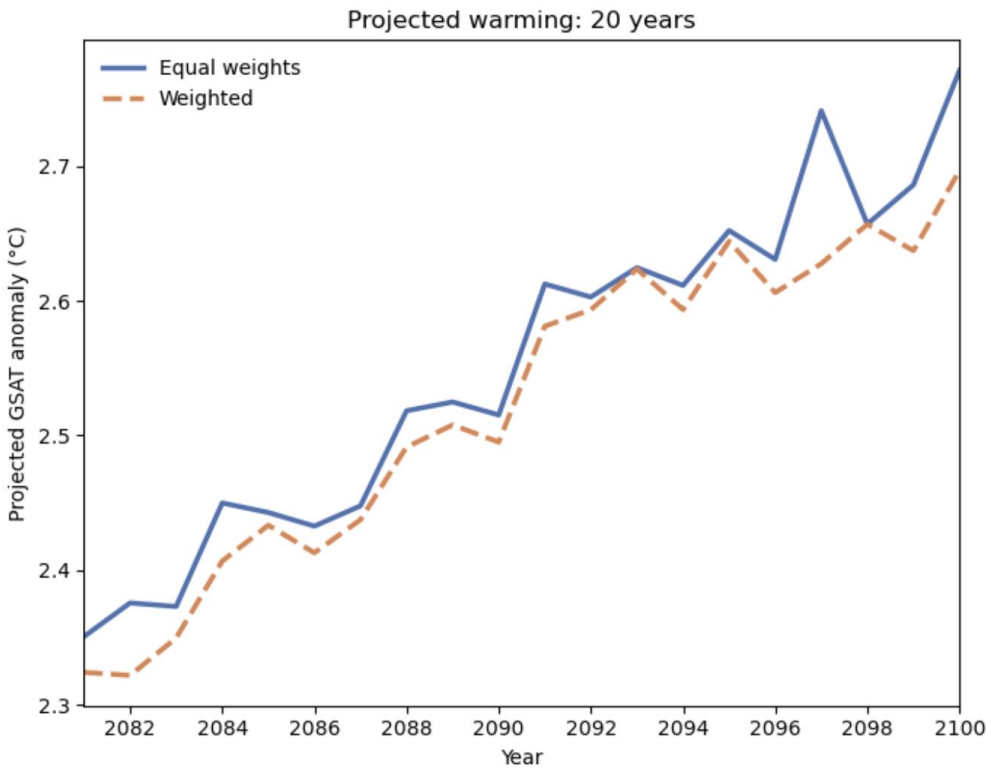


Why does the range become narrower?

- Weighted ensemble shows a narrower uncertainty range compared to equal weighting
- Main possible reasons:
  - Low-performance models receive **lower weights**
  - Redundant models are **down-weighted**
  - Ensemble is less influenced by outliers

➢ Weighting reduces ensemble spread and increases internal consistency

5. Results
2. Decrease in Projected Warming



Why does the temperature decrease?

- Weighted ensemble shows a **slight downward shift in median warming**
- Main possible reasons:
  - Models with high warming often have **higher RMSE → lower weight**
  - Models that are **closer to observations (lower RMSE)** tend to project more moderate warming

➢ Equal weighting may overestimate warming due to inclusion of low-skill models

6. Discussion

1. Limitations

1. Assumptions in weighting • RMSE & diagnostics choice is subjective • Assumes past performance → future skill • Limited physical interpretability	**3. Risk of overconfidence** • Narrower spread ≠ better uncertainty • Model filtering, not true uncertainty reduction
2. Model dependence not fully resolved • Similarity metric is incomplete • Shared model genealogy remains • Risk of redundant models	**4. Limited validation** • Hindcast performed, but limited in scope • Based mainly on historical GSAT

➢ **Weighting improves consistency, but may underestimate true uncertainty.**

6. Discussion

2. Future Work

1. Validation • Extend Hindcast experiment to more variables and scenarios • Evaluate uncertainty coverage？？	**3. Alternative weighting** • Bayesian frameworks • Emergent constraints • Machine learning
2. Robustness • Sensitivity to σ and diagnostics • Alternative performance metrics	**4. Broader evaluation** • Regional variables • Extremes & process-based metrics

➢ **Future work aims to improve robustness, interpretability, and predictive skill.**

Sources

- Figure 1: Reichler, T., & Kim, J. (2008). *How well do coupled models simulate today's climate?* **Bulletin of the American Meteorological Society, 89**(3), 303–311. https://doi.org/10.1175/BAMS-89-3-303
- Figure 2: Gleckler, P. J., Taylor, K. E., & Doutriaux, C. (2008). *Performance metrics for climate models*. Journal of Geophysical Research: Atmospheres, 113(D6), D06104. https://doi.org/10.1029/2007JD008972
- Figure 3: Knutti, R., Masson, D., & Gettelman, A. (2013). *Climate model genealogy: Generation CMIP5 and how we got there*. Geophysical Research Letters, 40(6), 1194–1199. https://doi.org/10.1002/grl.50256

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- Slide 6: Knutti, R., Sedláček, J., Sanderson, B. M., Lorenz, R., Fischer, E. M., & Eyring, V. (2017). *A climate model projection weighting scheme accounting for performance and interdependence*. **Geophysical Research Letters, 44**(4), 1909–1918. https://doi.org/10.1002/2016GL072012

Problems:

- Overstatement of improvements? - narrower range does not imply improved predictive skill rather just a tigther prediction?

=> why should I trust the weighted result more?

problems of different number of ensemble members

why does this model naturally handle multiple initial condition ensembles

σ_D and σ_S determines how strongly the model performance and similarity are weighted

WHY RMSE?

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KAI NOTES

Notes

(slide 23) to quantify uncertainty in a prediction we need to sample the space of plausible models

-> achieved by perturbing parameters of a single model

Ensemble member - simulation run of a climate model

weighting either of the singular ensemble members or of the mean of an model (mean over all members)

models - no singular best model - some models perform better than other at different

-> depends on the evaluation metric we are looking at

-> this is why we want to use all models

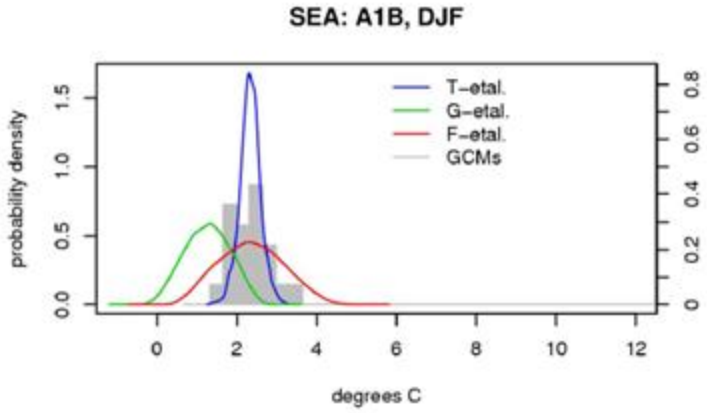
concept of a best model is illdefined

“Most models are strongly tied to their predecessors and some also exchange ideas and code with other models, thus supporting an earlier hypothesis that the models in the new ensemble are neither independent of each other nor independent of earlier generations” - Reto Knutti

Emergent constraint - way to evaluate our models to quantify relevant processes for climate system response

metric that includes a feedback and a temperature response - climate change parameter

An **emergent constraint** is a relationship that appears **across the model ensemble**:future warming=a+b·historical trend.

relationship between observables and predictions

get very different results based on the assumptions we make ->